



Carisurg

Assignment 8

Final Submission

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Problem Statement

Emergency departments in resource-constrained Caribbean settings like Mercer General Hospital operate on paper-based triage systems that generate no structured, queryable data. Without a digital foundation, locally validated AI triage tools cannot be built, and globally trained models — developed predominantly on North American and European patient populations — cannot be reliably applied. This project proposes to establish a structured digital triage dataset from the Mercer General ED as the necessary first step toward a clinically credible, regionally relevant AI-assisted triage classifier.

Previous Works & How We Move Forward

Research into AI and machine learning in ED triage has grown significantly, with studies showing promise in earlier diagnosis and intervention, though concerns remain around overconfident outputs presenting risks to patients (Tyler et al., 2024). AI-driven triage systems have demonstrated potential by automating patient prioritisation through real-time analysis of vital signs, medical history, and presenting symptoms, with narrative reviews covering literature from 2015 to 2024 identifying both benefits and persistent limitations (Da’Costa et al., 2025).

A systematic review of prospective studies found that while AI exhibits potential in emergency patient triage, it has not yet established evidence for substituting human involvement — a critical distinction for a setting like Mercer, where nurse-led triage is the primary model and clinician trust is a known adoption barrier (Yi et al., 2025).

Infrastructure constraints, including unreliable power supply, limited internet connectivity, and the absence of EHR integration tools, alongside a lack of high-quality annotated datasets for specific populations, represent key barriers to AI triage implementation in resource-limited settings. These barriers are directly applicable to the Caribbean context. A recent scoping review examining AI triage across low- and middle-income countries found that 72% of studies were conducted in high-income countries, with LMICs representing only 28% of the evidence base — reflecting both the thinness of the evidence base in resource-constrained settings and the near-complete absence of Caribbean data from this literature (Abdalhalim et al., 2025).

Araouchi & Adda (2024) demonstrated a multimodal AI triage system integrating structured vitals with unstructured clinical text, showing accuracy improvements over single-modality approaches — but validated exclusively on datasets from high-income settings with complete electronic health records, a precondition Mercer does not currently meet.

Where machine learning models have been developed and internally validated in non-Western settings, results are promising but context-dependent. Ueareekul et al. (2025) developed an XGBoost model across 163,452 ED visits at a Thai hospital, achieving superior predictive performance over the Canadian Triage and Acuity Scale (AUROC 0.917 vs. 0.882) — but noted that generalisability to different patient populations and healthcare infrastructures remains unproven, underscoring the need for locally trained models in every distinct clinical context.

Critically, there is precedent for the Phase 1 approach proposed here. Bhattarai et al. (2024) documented triage implementation at Vila Central Hospital in Vanuatu — a Pacific Small Island Developing State with infrastructure constraints directly analogous to the Caribbean — demonstrating the feasibility of a low-cost electronic data registry system built to capture triage

data in a previously paper-based ED. Their key finding was that a bespoke needs assessment of local triage requirements was essential before any digital tool could be designed, a lesson directly applicable to the Mercer context.

Proposed Solution

This project proposes a two-phase approach.

1. Phase 1 — the 12-week pilot — focuses on designing and validating a structured digital triage capture tool aligned with the existing Mercer General paper triage form, producing a clean, de-identified, queryable dataset from real ED encounters.
2. Phase 2, beyond this programme, would use that dataset to train and locally validate a rule-based ESI classifier, iterating toward a machine learning model as data volume grows.

The Phase 1 deliverable is not an AI model. It is the dataset that makes a future model possible — and as Dr De Fretias has noted, this alone is genuinely valuable. The tool would be designed for the nurse-led triage model at Mercer, with a minimal friction interface that respects the 3–5 minute triage window and does not increase documentation burden.

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